Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/05/2024
PI Name:	Smooke, Mitchell Ph.D.		
Project Title:	Computational and Experimental Analysis of SPICE Data Sets		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	New Haven	**State:**	CT
Zip Code:	06511-8940	**Congressional District:**	3
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	06/05/2019	**End Date:**	08/29/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Dietrich, Daniel L	**Contact Phone:**	216-433-8759
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/29/2022 per NSSC information; original end date was 6/4/2021 (Ed. 12/14/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Long, Marshall Ph.D. (Yale University)		
Grant/Contract No.:	80NSSC19K1634		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We plan to combine computational and experimental methods to generate temperature and soot volume fractions from the Smoke Point in Coflow Experiment (SPICE) data sets. A series of computations incorporating a soot model in which the dynamical equations for particle production are coupled to the flow and gaseous species conservation equations with modifications to account for the large soot loadings of the SPICE flames will be employed. Specifically, we will include the effects of thermal radiation re-absorption in the model and we will allow aggregate formation using a discrete-sectional size representation. For the heavy soot loading flames to be considered in this study, the balance between soot production, soot oxidation, and radiative temperature reduction will need to be examined more closely to help assess the various modes for smoke point development. To ascertain the importance of the various parameters in the model/kinetic mechanism, a first-order sensitivity analysis will be employed. The raw data from SPICE that we will concentrate on consists of more than 1600 images from the Nikon D300s DSLR camera, stored in a lossless, compressed format (Nikon .nef). These will be decoded to the three color channels and used to determine soot temperature from color ratio pyrometry. In the initial work, we will assume optically thin behavior, which will likely be accurate for some, but not all, of the fuels and flow conditions. The temperature distribution will then be used, along with the absolute light calibration that is available for the SPICE setup, to determine the soot volume fraction. This initial processing will provide valuable information on flame size, shape, and soot loading trends. This, along with the measured flow boundary conditions, will guide the computations. For the flames with the highest soot loading, the soot luminosity and spectral distribution will be affected by absorption and scattering. This in turn will affect the temperature and soot volume fraction calculated from the raw data. Taking optical properties of soot from similar flames in normal gravity, we will investigate the influence of extinction on the derived temperature and soot volume fraction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Compiled from final technical report submitted to NASA (Ed., 3/5/24): The global pandemic and COVID-19 issues made it difficult to keep the work going in a steady manner. Nevertheless, progress was made, and in the No Cost Extension (NCE) year, we obtained a much better understanding of the issues that control the large discrepancy between soot formation in 1g and in 0g flames. For some choices of model parameters, it proved possible to generate rather large soot loads, especially in microgravity and for lower values of the sectional coalescence cutoff. Unfortunately, however, the goal of replicating the significant discrepancies in peak soot when going from 1g to 0g remained elusive. Many uncertainties remain, and one in particular is already under investigation: we have focused on the surface growth model for surface-growth-dominated flames, but there could be interactions that we are missing with the soot inception model. At present, our soot model does not include large soot precursors such as pyrene; we plan to address this in future research. Results of this investigation were published in Combustion Science and Technology (see Cumulative Bibliography): Dobbins RR, Tinjero J, Squeo J, Zhao X, Hall RJ, Colket MB, Long MB, Smooke MD. A combined experimental and computational study of soot formation in normal and microgravity conditions. Combust. Sci. Technol. 2022 Mar 29. https://
Bibliography Type:	Description: (Last Updated: 03/05/2024)
Articles in Peer-reviewed Journals	Dobbins RR, Tinjero J, Squeo J, Zhao X, Hall RJ, Colket MB, Long MB, Smooke MD. "A combined experimental and computational study of soot formation in normal and microgravity conditions." Combust. Sci. Technol. 2022 Mar 29. https://doi.org/10.1080/00102202.2022.2041621 , Mar-2022